

Services vs Merchandise Trade

World Development Indicators (WDI) Analysis — 2000–2024

Research Question: Has the expansion of global transport, travel, and financial service trade outperformed physical merchandise trade growth from 2000 to 2024?

1. Objective / Problem Statement

The objective is to compare the long-term growth of service exports with merchandise exports from 2000 to 2024, while examining the composition of selected service exports—transport, travel, and insurance & financial services—and regional differences in service-export growth.

2. Dataset Description

The original WDI dataset contained 396,970 observations and 70 columns. The notebook selected five indicators relevant to the research question and restricted the analysis to 2000–2024.

Indicator	WDI Code	Unit
Service exports	BX.GSR.NFSV.CD	Current US\$
Goods exports	BX.GSR.MRCH.CD	Current US\$
Transport services	BX.GSR.TRAN.ZS	% of service exports
Travel services	BX.GSR.TRVL.ZS	% of service exports
Insurance & financial	BX.GSR.INSF.ZS	% of service exports

3. Data Cleaning Process

- Filtered the original WDI data to the five selected indicators.
- Restricted the analytical period to 2000–2024.
- Reshaped annual year columns into long format for time-series analysis.
- Calculated missing-value percentages for each selected indicator.
- Did not apply blanket imputation to missing values.

Indicator	Missing (%)
Service exports	16.38%
Goods exports	16.62%
Transport	18.49%
Travel	18.40%
Insurance & Financial	22.91%

4. Analysis Methodology

- **Annual growth:** $(\text{Value in year } t - \text{Value in previous year}) / \text{Value in previous year} \times 100$.
- **Total growth:** $(\text{Value in 2024} - \text{Value in 2000}) / \text{Value in 2000} \times 100$.
- **CAGR:** $(\text{Value in 2024} / \text{Value in 2000})^{(1/24)} - 1$.
- **Service-to-merchandise ratio:** Service exports / Goods exports.
- Regional comparison used service-export total growth and CAGR for the regional observations included in the notebook.
- Service composition was evaluated using transport, travel, and insurance & financial service indicators. The notebook also estimated monetary category values and calculated each selected category's share of the combined three-category total.

5. Key Findings

Services outpaced merchandise exports in CAGR terms. The notebook reports a 7.30% service-export CAGR compared with 6.34% for merchandise exports from 2000 to 2024. The difference is 0.96 percentage points.

Regional service growth varied. SAS had the highest reported regional service CAGR at 13.42%, followed by CEB at 9.15%. NAC had the lowest reported value among regions with valid results at 5.84%. MEA had missing start/end values in the notebook's regional calculation.

Region	CAGR
SAS	13.42%
CEB	9.15%
ECS	7.25%
AFE	7.24%
EAS	7.09%
SSF	7.06%
LCN	6.56%
AFW	6.47%
NAC	5.84%

6. Service Composition Findings

The following percentages represent the share of the **three selected service categories**—not the share of all service exports.

Period	Insurance & Financial	Transport	Travel
Pre-COVID	19.48%	32.78%	47.75%
COVID	29.56%	43.67%	26.77%
Post-COVID	21.70%	38.03%	40.28%

- **Travel dominated before COVID**, with a 47.75% average share of the three selected categories.
- **COVID materially altered the composition**: travel fell to 26.77%, while transport rose to 43.67% and insurance & financial services to 29.56%.
- **Travel recovered after COVID** to 40.28%, while transport moderated to 38.03% and insurance & financial services to 21.70%.
- In 2024, the selected-category composition was 43.53% travel, 34.99% transport, and 21.48% insurance & financial services.

7. Key Insights

- Service exports demonstrated stronger long-term growth than merchandise exports on a CAGR basis.
- Service trade is heterogeneous: travel, transport, and financial services responded differently to the COVID disruption.
- Travel was highly sensitive to the disruption and later showed substantial recovery.
- Transport became relatively more prominent during COVID, indicating the importance of logistics and connectivity.
- Insurance & financial services also gained relative importance during COVID before moderating post-COVID.
- Regional service-export performance varied substantially, creating opportunities for benchmarking.
- Missingness is material and should be considered when interpreting country, regional, and category comparisons.

8. Recommendations

- Diversify service exports across travel, transport, and financial services.
- Invest in resilient transport, logistics, digital, and cross-border infrastructure.
- Strengthen tourism resilience through market diversification, digital delivery, and crisis preparedness.
- Expand internationally tradable financial and digitally delivered services.
- Benchmark regions with stronger service-export growth and examine their enabling infrastructure, skills, connectivity, and policy environment.
- Improve data completeness and never interpret missing WDI observations as zero.
- Use the Power BI dashboard to monitor service CAGR, goods CAGR, annual growth, service-to-merchandise ratio, composition, and regional performance.

9. Conclusion

The analysis supports the conclusion that service exports expanded faster than merchandise exports over 2000–2024 in the notebook's reported results. Services recorded a 7.30% CAGR versus 6.34% for goods. The service composition analysis also shows that COVID-19 substantially changed the relative roles of travel, transport, and financial services, with travel subsequently recovering.

These findings suggest that service trade should be treated as an important component of trade-growth strategy, while resilience and diversification are necessary because individual service categories can respond differently to major disruptions.

10. Important Methodological Limitation

The notebook calculates global totals by summing observations by year. The WDI extract contains both individual countries and aggregate/region observations. If country and aggregate observations are summed together, some economic activity can be counted more than once. Therefore, the reported global CAGR figures are best treated as the result of the notebook's aggregation method. For a publication-quality global figure, the analysis should use a clearly defined country-only set or explicitly exclude aggregate regional observations.

Source: World Development Indicators (WDI), World Bank. Analysis based on `WDIanalysis.ipynb`.